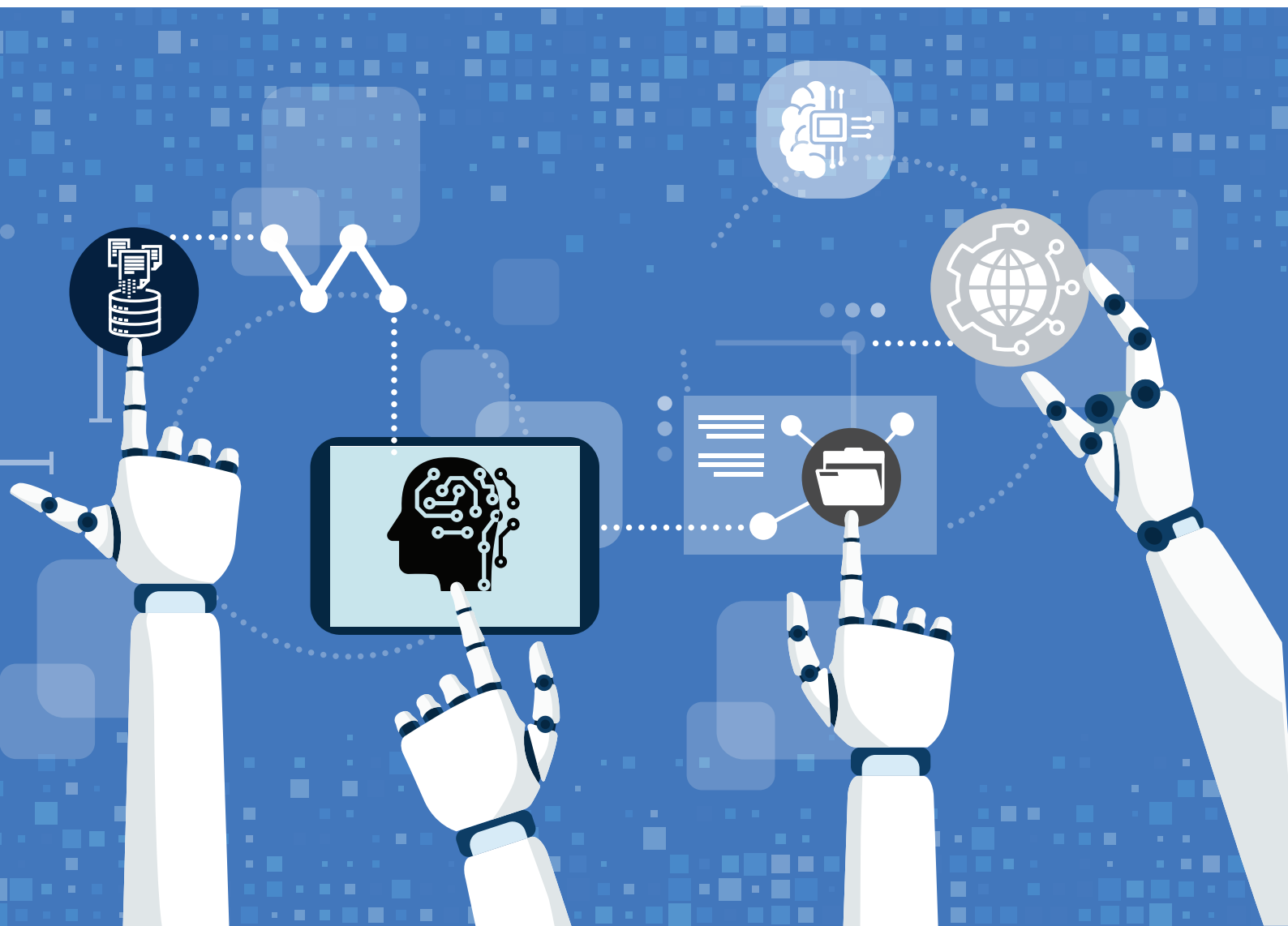


AI readiness for C-suite leaders



Preface

“AI readiness for C-suite leaders” is an MIT Technology Review Insights report sponsored by Fivetran. To produce this report, MIT Technology Review Insights conducted a global survey of C-suite and senior data executives across countries and industries. The report also draws on in-depth interviews conducted with business experts on data and AI.

Paul Kielstra was the author of the report, Teresa Elsey was the editor, and Nicola Crepaldi was the producer. The research is editorially independent, and the views expressed are those of MIT Technology Review Insights.

We would like to thank the following executives for their time and insights:

Stewart Bond, Vice President, Data Intelligence and Integration Software Service, IDC

George Fraser, CEO and Co-founder, Fivetran

Mike Hite, Chief Technology Officer, Saks

Suresh Venkatarayalu, Chief Technology and Innovation Officer, Honeywell

About the survey

The survey forming the basis of this report was conducted by MIT Technology Review Insights in February and March 2024. The survey sample consists of 300 C-suite executives and data and technology leaders. Eleven industries are represented: financial services, manufacturing, IT and telecommunications, consumer goods and retail, pharmaceutical and health care, government, travel and hospitality, professional services, energy and utilities, transport and logistics, and media and marketing.

Nearly all survey respondents (88%) come from the C-suite. These include chief executive officers (22%), chief information officers (23%), chief technology officers (21%), and chief data officers (10%). The respondents' organizations are headquartered in North America (30%); Europe, the Middle East, and Africa (27%); Asia-Pacific (27%); and Central and South America (17%). More than half of the respondents (53%) work at organizations with revenue of \$1 billion or more.

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01 Executive summary

Generative AI, like predictive AI before it, has rightly seized the attention of business executives. The technology has the potential to add trillions of dollars to annual global economic activity, and its adoption for business applications is expected to improve the top or bottom lines – or both – at many organizations.

While generative AI offers an impressive and powerful new set of capabilities, its business value is not a given. While some powerful foundational models are open to public use, these do not serve as a differentiator for those looking to get ahead of the competition and unlock AI's full potential. To gain those advantages, organizations must look to enhance AI models with their own data to create unique business insights and opportunities.

Preparing an organization's data for AI, however, unlocks a new set of challenges and opportunities. This MIT Technology Review Insights survey report investigates whether companies' data foundations are ready to garner benefits from generative AI, as well as the challenges of building the necessary data infrastructure for this technology. In doing so, it draws on insights from a survey of 300 C-suite executives and senior technology leaders, as well as on in-depth interviews with four leading experts.

Its key findings include the following:

Data integration is the leading priority for AI readiness. In our survey, 82% of C-suite and other senior executives agree that “scaling AI or generative

64% of C-suite leaders say data integration is a key investment area for AI and generative AI.

AI use cases to create business value is a top priority for our organization.” The number-one challenge in achieving that AI readiness, survey respondents say, is data integration and pipelines (45%). Asked about challenging aspects of data integration, respondents named four: managing data volume, moving data from on-premises to the cloud, enabling real-time access, and managing changes to data.

Executives are laser-focused on data management challenges – and lasting solutions. Among survey respondents, 83% say that their “organization has identified numerous sources of data that we must bring together in order to enable our AI initiatives.” Though data-dependent technologies of recent decades drove data integration and aggregation programs, these were typically tailored to specific use cases. Now, however, companies are looking for something more scalable and use-case agnostic: 82% of respondents are prioritizing solutions “that will continue to work in the future, regardless of other changes to our data strategy and partners.”

Data governance and security is a top concern for regulated sectors. Data governance and security concerns are the second most common data readiness challenge (cited by 44% of respondents). Respondents from highly regulated sectors were two to three times more likely to cite data governance and security as a concern, and chief data officers (CDOs) say this is a challenge at twice the rate of their C-suite peers. And our experts agree: Data governance and security should be addressed from the beginning of any AI strategy to ensure data is used and accessed properly.

02 Introduction: Data readiness drives AI value



Today's accelerating rollout of AI in all its forms, spurred by OpenAI's 2022's release of ChatGPT, is expected to reshape much of the economy. The consultancy McKinsey projects that, when fully deployed, predictive AI (including analytics and machine learning) will add between \$11.0 and \$17.7 trillion to annual global economic activity. Generative AI is projected to contribute an additional \$2.6 to \$4.4 trillion.¹

Companies are rushing to grab the opportunity. In our survey, 82% of C-suite and other senior executives agreed that "scaling AI or generative AI use cases to create business value is a top priority for our organization." These results are consistent with numerous other studies pointing to widespread AI adoption by companies across sectors and geographies.^{2,3,4}

Suresh Venkatarayalu, chief technology and innovation officer at Honeywell, says that his company has recently deployed generative AI and is already seeing a positive impact on productivity and efficiency.

Amid such widespread adoption, however, merely deploying open-source foundational AI models is unlikely to provide differentiation. Saks chief technology officer Mike Hite explains that "off-the-shelf" AI tools will give all users the same answers. Organizations that are pulling ahead do not necessarily have large AI teams or better large language model (LLM) technology. Instead, he says, "the people who have interesting data sets on a particular thing have opportunity."

Corralling data effectively

Executives seem to understand the strong link between data and effective AI strategies. CDOs are at the center of the deployment efforts, and almost all CDOs (97%) in our survey describe scaling AI as a top priority.

Organizations don't lack the data they need to meet that priority. Of those surveyed, 83% say that their organization "has identified numerous sources of data

